Stockholm University

Department of Meteorology

Numerical Methods in Atmospheric Sciences and Oceanography (MO8007)

# Home Assignment 4

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## Question

# Exercise 1 - Laplace Equation

Consider the Laplace equation:

$$\nabla^2 \Phi = 0$$

a) Discretise the equation in space.

b) Set $\Delta x = \Delta y$ and write the simplest iterative scheme (Jacobi iteration).

c) Set up a $4 \times 4$ grid, with an initial condition of $\Phi = 0$ and $\Phi = 1$ at the boundaries. Iterate 3 times the equation and enter all values.

d) Repeat the exercise using a faster iteration scheme (Gauss-Seidel or SOR).

# Exercise 2 - Semi-Implicit Schemes

Consider the non-rotating and $y$-independent shallow water equations on the form:

$$\begin{cases} \dfrac{\partial u}{\partial t} = -g \dfrac{\partial h}{\partial x}, & \text{(1a)} \\ \dfrac{\partial v}{\partial t} = 0, & \text{(1b)} \\ \dfrac{\partial h}{\partial t} = -H \dfrac{\partial u}{\partial x}. & \text{(1c)} \end{cases}$$

a) Discretise the equations (1a), (1b), (1c) using a semi-implicit scheme centred on time level $n$. Terms containing spatial partial derivatives must be evaluated as the average of the $(n-1)$th and $(n+1)$th time levels (discretise the equations on a C-grid). The implicitness parameter $\beta$ is set to 0.5.

b) Perform a stability analysis for the above discretised equations.

# Exercise 1 - Laplace Equation

**(a)** If one discretises the Laplace equation in space, one obtains:

$$\frac{\Phi_{i+1,j} - 2\Phi_{i,j} + \Phi_{i-1,j}}{(\Delta x)^2} + \frac{\Phi_{i,j+1} - 2\Phi_{i,j} + \Phi_{i,j-1}}{(\Delta y)^2} = 0 \tag{2}$$

**(b)** Setting $\Delta x = \Delta y$ and rearranging the above equation, one obtains

$$\Phi_{i,j} = \frac{1}{4} \left( \Phi_{i+1,j} + \Phi_{i-1,j} + \Phi_{i,j+1} + \Phi_{i,j-1} \right), \tag{3}$$

which simply is the four-point average of the surrounding points. To use the Jacobi iteration, one can write the above equation as

$$\Phi_{i,j}^{m+1} = \frac{1}{4} \left( \Phi_{i+1,j}^{m} + \Phi_{i-1,j}^{m} + \Phi_{i,j+1}^{m} + \Phi_{i,j-1}^{m} \right), \tag{4}$$

where the superscript $m$ denotes the iteration number.

**(c)** Using the above iterative scheme, one can set up a $4 \times 4$ grid with the initial condition of $\Phi = 0$ and $\Phi = 1$ at the boundaries. After 3 iterations, the grid values are then:

$$\begin{bmatrix} 1.000 & 1.000 & 1.000 & 1.000 \\ 1.000 & 0.000 & 0.000 & 1.000 \\ 1.000 & 0.000 & 0.000 & 1.000 \\ 1.000 & 1.000 & 1.000 & 1.000 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1.000 & 1.000 & 1.000 & 1.000 \\ 1.000 & 0.500 & 0.500 & 1.000 \\ 1.000 & 0.500 & 0.500 & 1.000 \\ 1.000 & 1.000 & 1.000 & 1.000 \end{bmatrix}$$

$$\Longrightarrow \begin{bmatrix} 1.000 & 1.000 & 1.000 & 1.000 \\ 1.000 & 0.750 & 0.750 & 1.000 \\ 1.000 & 0.750 & 0.750 & 1.000 \\ 1.000 & 1.000 & 1.000 & 1.000 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1.000 & 1.000 & 1.000 & 1.000 \\ 1.000 & 0.875 & 0.875 & 1.000 \\ 1.000 & 0.875 & 0.875 & 1.000 \\ 1.000 & 1.000 & 1.000 & 1.000 \end{bmatrix}$$

The values in the center of the grid thus obtained a value of 0.875 after 3 iterations.

**(d)** Another way to solve the Laplace equation is to use the Gauss-Seidel iteration, which is similar to the Jacobi iteration but uses the most recent values available. The Gauss-Seidel iteration can be written as

$$\Phi_{i,j}^{m+1} = \frac{1}{4}\left(\Phi_{i+1,j}^{m} + \Phi_{i-1,j}^{m+1} + \Phi_{i,j+1}^{m} + \Phi_{i,j-1}^{m+1}\right). \tag{5}$$

Using this iterations scheme one has to update the grid values from left to right and from top to bottom. After 3 iterations, the grid values are then:

$$\begin{bmatrix} 1.0000 & 1.0000 & 1.0000 & 1.0000 \\ 1.0000 & 0.0000 & 0.0000 & 1.0000 \\ 1.0000 & 0.0000 & 0.0000 & 1.0000 \\ 1.0000 & 1.0000 & 1.0000 & 1.0000 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 & 1.0000 \\ 1.0000 & 0.5000 & 0.6250 & 1.0000 \\ 1.0000 & 0.6250 & 0.8125 & 1.0000 \\ 1.0000 & 1.0000 & 1.0000 & 1.0000 \end{bmatrix}$$

$$\Longrightarrow \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 & 1.0000 \\ 1.0000 & 0.8125 & 0.9062 & 1.0000 \\ 1.0000 & 0.9062 & 0.9531 & 1.0000 \\ 1.0000 & 1.0000 & 1.0000 & 1.0000 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 & 1.0000 \\ 1.0000 & 0.9531 & 0.9766 & 1.0000 \\ 1.0000 & 0.9766 & 0.9883 & 1.0000 \\ 1.0000 & 1.0000 & 1.0000 & 1.0000 \end{bmatrix}$$

The values in the center of the grid thus resulted in values between 0.9531 and 0.9883 after 3 iterations. One can also compare this result with the Successive Over Relaxation (SOR) iteration scheme which uses

$$u_{i,j}^{m+1} = u_{i,j}^{m} + \omega c, \text{ where } c = u_{i,j}^{m+1} - u_{i,j}^{m} \tag{6}$$

for a specific relaxation factor $\omega$. The result for $\omega = 1.5$ gave

$$\begin{bmatrix} 1.0000 & 1.0000 & 1.0000 & 1.0000 \\ 1.0000 & 0.0000 & 0.0000 & 1.0000 \\ 1.0000 & 0.0000 & 0.0000 & 1.0000 \\ 1.0000 & 1.0000 & 1.0000 & 1.0000 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 & 1.0000 \\ 1.0000 & 0.7500 & 1.0312 & 1.0000 \\ 1.0000 & 1.0312 & 1.5234 & 1.0000 \\ 1.0000 & 1.0000 & 1.0000 & 1.0000 \end{bmatrix}$$

$$\Longrightarrow \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 & 1.0000 \\ 1.0000 & 1.1484 & 1.2363 & 1.0000 \\ 1.0000 & 1.2363 & 0.9155 & 1.0000 \\ 1.0000 & 1.0000 & 1.0000 & 1.0000 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1.0000 & 1.0000 & 1.0000 & 1.0000 \\ 1.0000 & 1.1030 & 0.8888 & 1.0000 \\ 1.0000 & 0.8888 & 0.9588 & 1.0000 \\ 1.0000 & 1.0000 & 1.0000 & 1.0000 \end{bmatrix}$$

The values in the center of the grid thus resulted in values between 0.8888 and 1.1030 after 3 iterations. However if the relaxation factor instead was set to $\omega = 1.1$ a better result of values between 1.0006 and 1.0010 was obtained.

**Answer**

The Jacobi iteration resulted in a homogenous convergence towards the final value of 1. After 3 iterations the Jacobi iteration on a 4x4-grid gave a homogenous result of 87.5% to the real analytical result. The Gauss-Seidel iteration resulted in a inhomogeneous convergence, but faster with a mean result of 97.365% to the real answer. The SOF iteration resulted in a mean 95.985% to the real answer. However using a relaxation factor of $\omega = 1.1$ resulted in a result of 100.085% to the real answer.

Thus the best answer was obtained from the SOF iteration with a clever choice of the relaxation factor, however, this might result in a small overestimation of the values. The Gauss-Seidel might otherwise be the best iteration, if homogenous convergence is not of the utter most importance.

# Exercise 2 - Semi-implicit schemes

**a** The grid used to discretize Equation 1 can be seen in Figure 1. Discretizing Equation 1a and Equation 1c (we do not discretize Equation 1b since it is trivial) we obtain:

$$\begin{cases} \dfrac{u_i^{n+1} - u_i^{n-1}}{2\Delta t} = -g\left[(1-\beta)\dfrac{h_{i+1}^{n+1} - h_i^{n+1}}{\Delta x} + \beta\dfrac{h_{i+1}^{n-1} - h_i^{n-1}}{\Delta x}\right] & \text{(7a)} \\ \dfrac{h_i^{n+1} - h_i^{n-1}}{2\Delta t} = -H\left[(1-\beta)\dfrac{u_i^{n+1} - u_{i-1}^{n+1}}{\Delta x} + \beta\dfrac{u_i^{n-1} - u_{i-1}^{n-1}}{\Delta x}\right], & \text{(7b)} \end{cases}$$

where we know that the implicit parameter is $\beta = 0.5$.

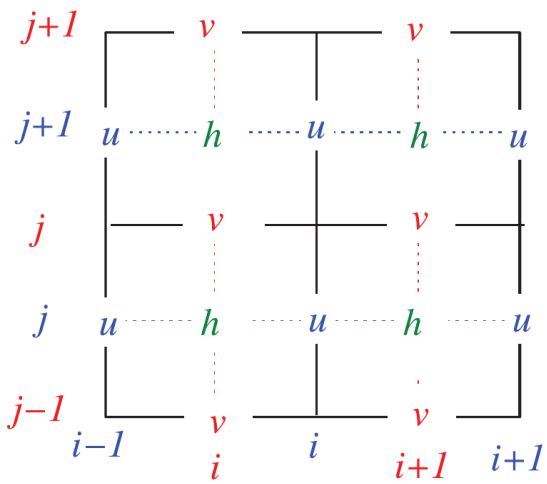


**Figure 1:** This figure displays the Arakawa c-grid.

**(b)** To evaluate the stability Equation 7 one has to use the Von Neumann method as

$$\begin{pmatrix} u_i^n \\ h_i^n \end{pmatrix} = \begin{pmatrix} u_0 \\ h_0 \end{pmatrix} e^{\mathfrak{I}(ik\Delta x - \omega_D n\Delta t)} \implies \begin{matrix} u_i^{n\pm1} = \lambda^{\pm1} u_i^n \\ u_{i\pm1}^n = e^{\pm\mathfrak{I}k\Delta x} u_i^n \end{matrix} \quad \text{and} \quad \begin{matrix} h_i^{n\pm1} = \lambda^{\pm1} h_i^n \\ h_{i\pm1}^n = e^{\pm\mathfrak{I}k\Delta x} h_i^n \end{matrix}$$

where $\mathfrak{I}$ is the imaginary unit, $k$ is the wavenumber nad $\lambda$ is the relative step size in time.

This transforms Equation 7 into

$$\begin{aligned} \left(\lambda - \lambda^{-1}\right) u_i^n &= -\frac{g\Delta t}{\Delta x}\left(\lambda e^{\mathfrak{I}k\Delta x} - \lambda + \lambda^{-1} e^{\mathfrak{I}k\Delta x} - \lambda^{-1}\right) h_i^n \\ \left(\lambda - \lambda^{-1}\right) h_i^n &= -\frac{H\Delta t}{\Delta x}\left(\lambda - \lambda e^{-\mathfrak{I}k\Delta x} + \lambda^{-1} - \lambda^{-1} e^{-\mathfrak{I}k\Delta x}\right) u_i^n. \end{aligned}$$

Combining the two equations above and using the Courant number $\mu = \frac{c\Delta t}{\Delta x}$, where $c = \sqrt{gH}$ gives

$$\begin{aligned} \left(\lambda - \lambda^{-1}\right)^2 &= \mu^2\left(\lambda e^{\mathfrak{I}k\Delta x} - \lambda + \lambda^{-1} e^{\mathfrak{I}k\Delta x} - \lambda^{-1}\right)\left(\lambda - \lambda e^{-\mathfrak{I}k\Delta x} + \lambda^{-1} - \lambda^{-1} e^{-\mathfrak{I}k\Delta x}\right) \\ &= \mu^2\left(\lambda^2 e^{\mathfrak{I}k\Delta x} - \lambda^2 + e^{\mathfrak{I}k\Delta x} - 1 - \lambda^2 + \lambda^2 e^{-\mathfrak{I}k\Delta x} - 1 + e^{-\mathfrak{I}k\Delta x}\right. \\ &\qquad \left. e^{\mathfrak{I}k\Delta x} - 1 + \lambda^{-2} e^{\mathfrak{I}k\Delta x} - \lambda^{-2} - 1 + e^{\mathfrak{I}k\Delta x} - \lambda^{-2} + \lambda^{-2} e^{-\mathfrak{I}k\Delta x}\right). \\ &= \mu^2\left[\lambda^2\left(2\cos(k\Delta x) - 2\right) + 2\left(2\cos(k\Delta x) - 2\right) + \lambda^{-2}\left(2\cos(k\Delta x) - 2\right)\right] \\ &= 2\mu^2\left[\cos(k\Delta x) - 1\right]\left(\lambda^2 + 2 + \lambda^{-2}\right) \\ &= 2\mu^2\left[\cos(k\Delta x) - 1\right]\left(\lambda + \lambda^{-1}\right)^2. \end{aligned}$$

Now one can use the fact that $\left(\lambda - \lambda^{-1}\right)^2 = \left(\lambda + \lambda^{-1}\right)^2 - 4$ together with the fact that $1 - \cos(2\theta) = 2\sin^2(\theta)$ and obtain

$$\begin{aligned}
\frac{\left(\lambda + \lambda^{-1}\right)^2 - 4}{\left(\lambda + \lambda^{-1}\right)^2} &= -4\mu^2 \sin^2\left(\frac{k\Delta x}{2}\right) \\
1 - \frac{4}{\left(\lambda + \lambda^{-1}\right)^2} &= -4\mu^2 \sin^2\left(\frac{k\Delta x}{2}\right) \\
\left(\lambda + \lambda^{-1}\right)^2 &= \frac{4}{1 + 4\mu^2 \sin^2\left(\frac{k\Delta x}{2}\right)} = B^2.
\end{aligned}$$

Where $B$ is a real positive number in the range $B \in \left[2\sqrt{\frac{1}{1+4\mu^2}}, 2\right]$, where $0 < \sqrt{\frac{1}{1+4\mu^2}} < 1$. One can thus also state that $B \in (0, 2]$ Solving this second order equation in $\lambda$ gives:

$$\lambda^2 - B\lambda + 1 = 0 \implies \lambda_{1,2} = \frac{B}{2} \pm \sqrt{\frac{B^2}{4} - 1}$$

Now we obtain two cases, where the root is imaginary or real.

- If $\frac{B^2}{4} - 1 < 0$ we have an imaginary solution: This would yield that

  $$\lambda_{1,2} = \frac{B}{2} \pm \Im\sqrt{1 - \frac{B^2}{4}}.$$

  We thus obtain a norm of

  $$\lambda^2 = \frac{B^2}{4} + 1 - \frac{B^2}{4} = 1.$$

  Thus if $\frac{B^2}{4} - 1 < 0$ (or $B^2 < 4$) then we have an unconditionally stable solution!

- The real solution occurs when $\frac{B^2}{4} - 1 > 0$, or in other words $B^2 \geq 4$. One can note that we have the limit $B^2$ cannot exceed the value of 2, but can obtain the value when the sin-term goes to zero. Therefore this occurs if and only if $B = 2$, which would yield

  $$\lambda = 1 \pm 0 = 1,$$

  which also is unconditionally stable!

**Answer**

The scheme is unconditionally stable for any choice of Courant number. However, the solution does obtain imaginary time steps which indicate oscillatory behaviour. Furthermore, the oscillatory behaviour is independent of the of Courant number, and will thus always exist. However this is probably due to the nature of the shallow water equations having a wave solution. Observing the numerical solution in Figure 2, one can see normal behaviour the solution.

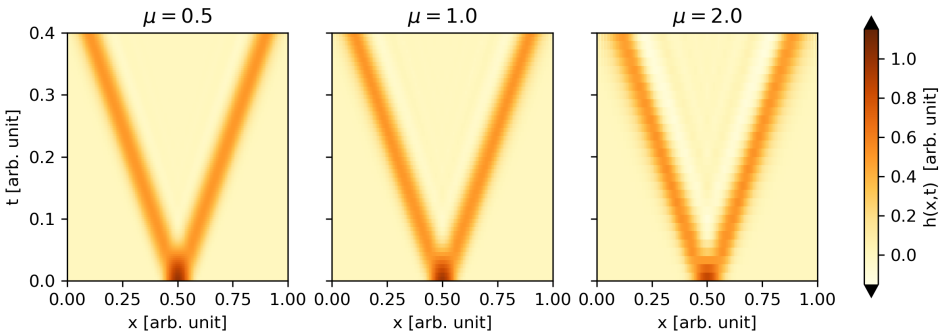


**Figure 2:** This figure displays the real solution when $g = 1$ and $H = 1$, and the initial condition is a localised cosine-pulse. The solution was obtained through the use of the matrix-method.